Sayak Chatterjee

Curriculum Vitae

Personal Details

Name Sayak Chatterjee

Nationality Indian

Date of Birth 18th December, 1995

Sex Male

Email Id sayakchatter@umass.edu, sayak@jlab.org, sayakchatterjee896@gmail.com

Designation Postdoctoral Researcher at University of Massachusetts, Amherst, USA

Education

2018 - 2023 **Ph.D. in Physics (Experimental High Energy Nuclear Physics)**
University University of Calcutta **Institution** Bose Institute, Kolkata, India
Thesis title Performance studies of Gas Electron Multiplier detector for the Muon Chamber of high rate CBM experiment at FAIR

2016 - 2018 **M. Sc. in Physics - Focus on detector R&D for high rate nuclear physics experiments**
University University of Calcutta **Institution** Bose Institute, Kolkata, India
Percentage of Marks 80.28 % **Class** 1st class **Year of passing** 2018

2013 - 2016 **Bachelor of Science (Physics Honours with Mathematics & Statistics)**
University University of Calcutta **Institution** St. Paul's Cathedral Mission College, Kolkata
Percentage of Marks 81.37 % **Class** 1st class **Rank** 2nd in the University
Year of passing 2016

2011 - 2013 **Higher Secondary** $(+2)$
Board West Bengal Council of Higher Secondary Education
Institution Sri Ramkrishna Sikshalaya, Howrah, West Bengal, India
Percentage of Marks 78.4 % **Year of passing** 2013

2011 **Secondary Examination**
Board West Bengal Board of Secondary Education
Institution Sri Ramkrishna Sikshalaya, Howrah, West Bengal, India
Percentage of Marks 74.5 % **Year of passing** 2011

Work Experience

2023 Nov - Current **Postdoctoral Researcher**
University University of Massachusetts, Amherst, USA
Collaboration MOLLER experiment at Thomas Jefferson Lab, Virginia, USA

2022 Dec - 2023 October **Research Associate**
University University of Massachusetts, Amherst, USA
Collaboration MOLLER experiment at Thomas Jefferson Lab, Virginia, USA

Research Interests

I did my Ph.D in Experimental High Energy Nuclear Physics. I have worked on the development of the Gas Electron Multiplier (GEM) detectors for the Compressed Baryonic Matter (CBM) experiment at GSI, Germany. I am inspired by the notion that the CBM experiment will explore the high density region of the QCD phase diagram to study the Quark Gluon Plasma; a new state of matter. I have investigated the long term stability, charging-up effect, aging and spark probability of triple GEM detectors. I have also done basic characteristics study of other gaseous detectors such as Resistive Plate Chamber (RPC), Straw Tube and also of plastic scintillators. I have also worked on the Monte Carlo based simulation studies to investigate the feasibility of di-muon detection at CBM experiment using the Muon Chamber (MuCh) sub-system using GEANT4. This work includes the optimisation of the geometry of the MuCh absorber and detector system and also reconstruction of the muon pairs coming from the decay of Low Mass Vector Meson (LMVM) and Charmonium (J/ψ). The detection of J/ψ is one of the important goals of the CBM experiment because no data is available in the heavy ion sector below the top SPS energy of 158 GeV. I have also worked on the phenomenological predictions of the Cold Nuclear Matter (CNM) effect on the J/ψ yield at CBM and SPS energies.

Here is the brief summary of the work I have done for the CBM experiment at GSI, Germany during my PhD period;

- Designed and built Gas Electron Multiplier (GEM) detectors for the Muon Chamber in the CBM experiment at FAIR
- Investigated long-term stability, charging-up effects, and spark probability of triple GEM detectors under high radiation conditions
- Conducted Monte Carlo-based simulations using GEANT4 for di-muon detection efficiency and physics performance studies
- Participated in beam test campaign at GSI, Germany to characterize GEM detector performance
- Phenomenological studies on Cold Nuclear Matter (CNM) effects on J/ψ production at FAIR and SPS energies
- Characterized Resistive Plate Chamber (RPC), Straw-tube detector for CBM experiment and Scintillators for cosmic muon studies

Currently I am working as a postdoctoral researcher in the MOLLER collaboration, which aims to measure the parity violating asymmetry in electron-electron (Møller) scattering with unprecedented precision. The MOLLER experiment aims to measure the electroweak mixing angle, a fundamental parameter of the electroweak theory, with unprecedented precision. The measurement will be carried out at Thomas Jefferson Laboratory, USA. I am involved in the characterisation and optimisation of the Cherenkov detectors proposed to be used for measuring the flux of scattered electrons. I am also involved in the GEANT4 based simulation to optimise the geometry of the detector sub-systems, development and testing of the Data Acquisition System and developing analysis framework for MOLLER experiment.

Here is the brief summary of the work I have been doing as a part of the MOLLER collaboration at Jefferson Lab, Virginia, USA;

- Leading the effort of characterizing production electron detection modules (200 detectors) for the final experiment
- Developing simulation framework and performance modelling of Cherenkov detectors, optimizing design using GEANT4 and CERN-ROOT (C++)
- Designing and implementing data analysis tools to evaluate detector precision and system noise
- Mentoring graduate and undergraduate students in the collaboration
- Data acquisition system development including implementation and testing of VME, MPD/APV25, fADC, QDC based readout systems
- Offline data analysis framework development using CERN-ROOT and Jefferson Lab software framework
- Presenting research findings at Jefferson lab collaboration meetings and at international conferences

Scholarships/Awards/Recognitions

- Nominated by the MOLLER collaboration to present the MOLLER overview in Jefferson Lab Hall A collaboration meeting, Jefferson Lab, Virginia, USA, January 2024.
- Received the award for best oral presentation in the Students' Symposium "Recent Trends in Natural Sciences, 2022" at Bose Institute, Kolkata, November 28-29, 2022.
- Invited to adjudicate the Senior School Science Exhibition at La Martiniere for Boys, Kolkata, India, 22nd July 2022
- Received the Financial support to attend the XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (QM2022), Krakow, Poland April 4-10, 2022
- Received the Young Research Grant to attend the 15th Pisa Meeting on Advanced Detector, La Biodola, Isola d'Elba, Italy, May 22-28, 2022
- Received the Ernest Rutherford best researcher award 2022 on 26.01.2022 by the International Multidisciplinary Research Foundation (IMRF)
- Selected as the only Indian student to deliver the Highlight talk on CBM-MuCh simulation at the plenary session of 37th CBM Collaboration Meeting (Online), 5th March 2021
- Selected as a participant in the 6th International FAIR School, Italy 7-14 September, 2019
- Best poster award at Workshop on Dynamics of QCD Matter, National Institute of Science Education and Research (NISER), Odisha, India, August 2019
- Selected as the only Indian student in the HGS-HIRe Summer Student program at GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany. 23^{rd}July - 13 th September, 2018
- Secured an all India rank 33 (Lectureship) in Joint CSIR-UGC Junior Research Fellowship & Eligibility for Lectureship (NET) Exam, June 2018
- Secured an all India rank 273 in Graduate Aptitude Test in Engineering GATE (Physics) 2018
- Ph.D. research fellowship, Bose Institute (2018 - 2022)
- Bose Institute fellowship to pursue M.Sc. in Physics (2016 - 2018)

- Secured the 2nd rank at the University of Calcutta in B.Sc. Physics Honours (2016)

Professional stays (not conferences)

GSI, Darmstadt, Germany (2019, 1 month)
Institute for Nuclear Physics, University of Mainz, Mainz, Germany (2023, 1 month)
Thomas Jefferson Lab (Jlab), Virginia, USA (2023, 2024, 2025)

Conferences & Schools attended

- Frontier Detectors for Frontier Physics 16th Pisa Meeting on Advanced Detectors, La Biodola, Isola d'Elba, Italy, May 26-June 1, 2024
- Frontier Detectors for Frontier Physics 15th Pisa Meeting on Advanced Detectors, La Biodola, Isola d'Elba, Italy, 22-28 May, 2022
- XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions, (QM2022) (Online), Krakow, Poland, April 4-10, 2022
- 65th DAE Nuclear Physics Symposium (Online), 1st December - 5th December, 2021
- 3rd National Conference on Frontiers of Modern Physics (NCFMP 2021) (Online), 26th November - 27th November, 2021
- IV ALICE-India School on Quark-Gluon Plasma (Online), 8th November - 20th November, 2021
- Workshop on Advanced Radiation Detector and Instrumentation in Nuclear and Particle Physics (RAPID2021) (Online), 25th October - 29th October, 2021
- VIRTUAL SCHOOL ON FLAVOR STRUCTURE OF THE STANDARD MODEL (Online), 31st August - 12th September, 2021
- CERN-Fermilab HCP Summer School (Online), 23rd August - 4th September, 2021
- RHIC Beam Energy Scan and Beyond (Online), 16-18 August, 2021
- Software Tools and Techniques used in EHEP and its Applications (Online), MNIT Jaipur, July 12 - 19, 2021
- Machine Learning for Particle Physics, Mainz Institute for Theoretical Physics (Online), Johannes Gutenberg University, 21st June - 2nd July, 2021
- 19th International Conference on Strangeness in Quark Matter (Online), Brookhaven National Laboratory, New York, 17-22 May, 2021
- Participated in the data taking of the mCBM campaign at GSI, Darmstadt, Germany, 7th November - 7th December 2019
- 6th International FAIR School, Italy, 7-14 September, 2019
- Workshop on Dynamics of QCD Matter, National Institute of Science Education and Research (NISER), Odisha, India, 15-17 August 2019
- International Workshop on Forward and Jet Physics at LHC, Bose Institute, Kolkata, 11-12 February 2018
- HGS-HIRe Summer Student program at GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany, 23^{rd}July - 13 thSeptember, 2018
- C.K. Majumdar Memorial Summer Workshop in Physics at S.N.Bose National Centre for Basic Sciences

(SNBNCBS), Salt Lake, Kolkata, India, 14-24 July, 2016

Computational Skills

Computer Languages	C++, C, Python, FORTRAN90	**Graph Plotters**	QTI plot, ROOT(CERN)
Event Generators	UrQMD, PLUTO	**Simulation Softwares**	GEANT, Garfield
Miscellaneous	Autodesk, Inventor, JT2Go		

Presentations

Oral

- Amherst Center For Fundamental Interactions (ACFI) seminar series, University of Massachusetts, Amherst (10th December 2024)
  Title of talk: Precision Test of Electroweak Physics: The MOLLER Experiment
- University of Massachusetts, Amherst IPO Scholar Spotlight Lecture (21st November 2024)
  Title of talk: Decoding the Invisible: The Intriguing World of Subatomic Particles
- MOLLER Collaboration meeting, Jefferson Lab (20-22th May 2023)
  Title of talk: MAINZ test beam: September 2023
- JLab Hall A Collaboration meeting, Jefferson Lab (16-17th Jan 2024)
  Title of talk: Overview of the MOLLER experimental apparatus
- MOLLER Simulation Computing Workshop, (26th May 2023)
  Title of talk: Simulation Computing Workshop: Hands-On Session
- MOLLER Collaboration meeting, Jefferson Lab (6th May 2023)
  Title of talk: R&D activities at UMass
- Recent Trends in Natural Sciences 2022, Bose Institute Students' Symposium (Nov 28-29)
  Title of talk: Performance Studies of Gas Electron Multiplier Detector for the Muon Chamber of High Rate CBM Experiment at FAIR
- 40th CBM Collaboration Meeting, 10-14 October 2022 (CBM Dilepton PWG)
  Title of talk: J/ψ yields in low energy nuclear collisions at FAIR: a baseline estimation (Online)
- 39th CBM Collaboration Meeting, 16-20 May 2022 (CBM Dilepton PWG)
  Title of talk: Beam Energy Scan via di-muon channel at SIS100 (Online)
- 3rd National Conference on Frontiers of Modern Physics (NCFMP 2021),26-27th November 2021
  Title of talk: Characterization of Gas Electron Multiplier Detector (Online)
- RD51 collaboration meeting, 16th November 2021
  Title of talk: Charging up effect in triple GEM detector (Online)
- Workshop on Advanced Radiation Detector and Instrumentation in Nuclear and Particle Physics (RAPID2021), 27th October 2021
  Title of talk: Gain, energy resolution measurement of GEM detector and coincidence setup demonstration using cosmic ray (Online)
- Workshop on Advanced Radiation Detector and Instrumentation in Nuclear and Particle Physics (RAPID2021), 26th October 2021
  Title of talk: Charging up effect in triple GEM detector (Online)

- 38th CBM Collaboration Meeting, 30th September 2021 (CBM Dilepton PWG)
  Title of talk: Reconstruction of LMVM cocktail with GEANT4 transport engine @ 12 A GeV/c (Online)
- VIRTUAL SCHOOL ON FLAVOR STRUCTURE OF THE STANDARD MODEL, 11th September, 2021
  Title of talk: Characterisation of Gas Electron Multiplier detector for high rate experiments
- 37th CBM Collaboration Meeting, 5th March 2021 (Highlight talk at the Plenary session)
  Title of talk: Physics performance studies with MuCh setup (Online)
- 37th CBM Collaboration Meeting, 3rd March 2021 (CBM Dilepton PWG)
  Title of talk: Reconstruction of LMVM with MuCh start version (Online)
- XXIV DAE-BRNS HIGH ENERGY PHYSICS SYMPOSIUM, 17th December 2020
  Title of talk: Feasibility studies of J/ψ measurement with CBM detector setup at FAIR SIS100 energies (Online)
- 36th CBM Collaboration Meeting, 20th October 2020 (CBM Dilepton PWG)
  Title of talk: Reconstruction of dimuon cocktail at FAIR SIS100: GEANT3 vs GEANT4 (Online)
- 35th CBM Collaboration Meeting, 24th March 2020 (MUCH Session)
  Title of talk: Optimization of the MUCH absorber configuration (Online)
- 35th CBM Collaboration Meeting, 24th March 2020 (CBM Dileptons PWG)
  Title of talk: Study of J/ψ production at SIS100 energies (Online)
- 34th CBM Collaboration Meeting, 29th September 2019 (MUCH session)
  Title of talk: Spark probability measurement of Single Mask GEM detector for CBM
- Vigyan Samagam, Mumbai & Bengaluru, June & September 2019
  Title of talk: Evolution of gaseous detectors for High Energy Physics experiments

Poster

- Frontier Detectors for Frontier Physics 16th Pisa Meeting on Advanced Detectors, May 2024
  Title of poster: Characterisation of Cherenkov detectors for the MOLLER experiment
- Frontier Detectors for Frontier Physics 15th Pisa Meeting on Advanced Detectors, 27th May 2022
  Title of poster: Visual investigation of possible degradation in GEM foil under test GEM chamber
- Frontier Detectors for Frontier Physics 15th Pisa Meeting on Advanced Detectors, 27th May 2022
  Title of poster: Effect of relative humidity on the long-term operation of a single mask triple GEM chamber
- XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (QM2022), 6th April 2022
  Title of poster: Prospect of dilepton and charmonium measurement with CBM experiment (Online)
- CERN-Fermilab HCP Summer School, 28rd August, 2021
  Title of poster: Effects of charging-up on Gas Electron Multiplier detector (Online)
- Workshop on Dynamics of QCD matter, NISER, 15-17 August 2019
  Title of poster: Characterisation of GEM detector for CBM muon chamber

Project students

- Mr. Bryant Wu, University of Massachusetts, Amherst, Summer Intern of 2024
  Project: Fabrication of plastic scintillators and their characterization

- Ms. Joelle Beck, University of Massachusetts, Amherst, Summer Intern of 2023
  Project: Development of a GEM based cosmic muon tracking system for characterizing Cherenkov detectors
- Mr. John Paul Anderson, University of Massachusetts, Amherst, Summer Intern of 2023
  Project: Software development for QDC based readout system
- Mr. K Yabe, University of Massachusetts, Amherst, Summer Intern of 2023
  Project: PMT calibration using different techniques
- Mr. Shreesh Sahai, Amity university, Summer Intern at Bose Institute 2022
  Project: Investigation of the instability in performance of triple GEM chamber
- Mr. Rajat Paul, Savitribai Phule Pune University, Summer Intern at Bose Institute 2022
  Title of the project: Characterisation of GEM detector

Research Experience

I completed my M.Sc. from Bose Institute under the Integrated M.Sc-PhD program in 2018 and joined as a Junior Research Fellow at the Department of Physics, Bose Institute under the supervision of Dr. Saikat Biswas. I have also done my master's dissertation under the guidance of Dr. Saikat Biswas. The title of my master's thesis is "Study of characteristics of triple GEM Detector for CBM experiment". During the break of 2nd semester of the master course, I have done a project under Dr. Saikat Biswas on the measurement of cosmic rays using plastic scintillator detectors. The outcome of the project is published as a proceeding of the DAE symposium on Nuclear Physics, 2017. At the end of my master's course, I have selected in the HGS-HIRe summer student program at GSI, Darmstadt, Germany and went to attain the school from 23rd July to 13th September 2018. I was the only Indian participant in that summer school. During the summer school, I worked on the optimization of the tracking algorithm and study of the dependency of mass resolution on target thickness and detector setup for the R^3B experiment.

I started working under the supervision of Dr. Saikat Biswas as a junior research fellow in August 2018. My primary responsibility was to study the performance of the triple Gas Electron Multiplier (GEM) chambers for the Muon Chamber (MuCh) at the Compressed Baryonic Matter (CBM) experiment. The goal of the CBM experiment is to explore the QCD (Quantum Chromo Dynamics) phase diagram at low temperature and moderate to high baryon densities. The MuCh sub-system at the CBM experiment will be used to track and identify the di-muons coming from the decay of low mass vector mesons (LMVM) and J/ψ mesons which carry the information about the medium created after the collision of heavy ions at relativistic energies. The GEM detectors will be used as the tracking detector for the MuCh detector sub-system.

To investigate the performance of the triple GEM chambers, the work is initiated by performing the quality assurance (QA) test of the GEM foils. The QA test includes the optical scanning and leakage test of the GEM foils. The assembly of the triple GEM chamber is also done at the laboratory of Bose Institute. The basic characterisation of the triple GEM prototype is carried out using Fe^{55} X-ray source (5.9 keV) and with Ar/CO_2 gas mixture in continuous flow mode. The same source is used to irradiate the chamber and also to record the 5.9 keV X-ray spectra. The X-ray spectra are stored using the ORTEC MCA and the data are analysed using the CERN-ROOT framework to get the gain and energy resolution of the chamber. The characterisation studies are carried out for both Double Mask (DM) and Single Mask (SM) triple GEM chambers. The uniformity in characteristics in terms of gain, energy resolution and count rate of the GEM chamber prototypes are investigated over the active area of the chambers and the results are reported in several articles. The investigation of the uniformity in performance is an essential criterion for any detector, used as a tracking device. The presence of the dielectric medium (Kapton)

inside the active medium of the chamber changes its behaviour and this phenomenon is known as the charging-up effect. To understand the charging-up phenomena, the chambers are irradiated at different gain and also with different irradiation rates using collimators of different dimensions. It is found that with increasing particle density, the charging-up time is reduced as expected. The typical charging-up time obtained at a gas gain of ~ 10^4 is ~ 20 minutes. The effect of the charging-up effect on the energy resolution of the chamber is also investigated. The long-term stability of the detector is investigated by placing the Fe^{55} source on a particular position of the chamber for a few months. During that time the energy spectra and the ambient temperature and pressure are recorded continuously. No significant degradation in the performance of the chamber is observed even after accumulating ~ 7 mC/mm^2 of charge. To establish the fact that the GEM chambers can be used in high-rate experiments, the spark probability of the detector is one of the important parameters that should be measured. A triple GEM prototype is tested at the CERN SPS beam line facility with a pion beam of momentum ~ 150 GeV/c. To mimic the situation of particle shower, an iron block is also used in the beam line. The spark in the chamber is identified by studying the drop in the GEM count rate as compared to the trigger counters. The spark probability of the chamber is found to be ~ 10^{-10} at a typical gas gain of ~ 5000 with the pion beam. No spark is observed with the shower setup and the probable reason for that is the reduced particle density over the active area of the chamber. I have received the Young Research Grant to attend and present my work on the performance studies of the triple GEM chamber at the 15th Pisa meeting on Advanced Detector, Italy, 2022. I have also participated in the mini-CBM beam time campaign at GSI, Germany in 2019 to study the performance of the large size triple GEM chambers in the nucleus-nucleus collisions and with the miniaturized version of the CBM detector sub-systems and data acquisitions systems.

In the above mentioned R&D work, I am involved in assembling detector modules, instrumenting them during the beam time, the designing of the experimental setup, data analysis using CERN-ROOT and also in the preparing of the manuscripts with the obtained results.

Apart from the R&D on the GEM detector, I was also actively involved in the fabrication of the Resistive Plate Chambers (RPC) for the last two stations of CBM-MuCh and the basic characterisation studies of scintillator detectors (plastic and NaI), straw-tube detectors and RPC detectors using radio-active sources and cosmic muons.

Along with the detector R&D, I was also involved in the Monte Carlo (MC) based feasibility studies of di-muon detection at CBM experiment using MuCh detector sub-system. I was one of the responsible persons for studying the different critical aspects of the realistic design simulation of the MuCh sub-system. This includes the finalization of the detector chamber geometry, absorber configurations, and choice of suitable detector configurations. The results of the investigations are reported as collaboration progress reports and internal notes. I was also involved in the physics performance studies of di-muon detection. I have studied the reconstruction performance of the freeze-out cocktail via the di-muon channel at the energies foreseen in the CBM experiment. The outcomes of these studies are presented as a poster by me on behalf of the CBM Collaboration at the Quark Matter 2022 conference. I have also studied the reconstruction performance of the J/ψ via the di-muon channel and presented on behalf of the CBM Collaboration at the XXIV DAE-BRNS High Energy Physics Symposium, 2020. I have been selected by the collaboration to give a highlight talk on CBM-MuCh simulation at the CBM Collaboration meeting 2021. I have also received the CBM Juniors Travel Grant to attend the Quark Matter 2022. I have done some phenomenological work to predict the J/ψ production at CBM and SPS energies.

After submitting my thesis, I have joined the Department of Physics at the University of Massachusetts, Amherst, USA on December 2022 as a research associate in Prof. Krishna Kumar's group to work in the

MOLLER (Measurement Of a Lepton Lepton Electroweak Reaction) collaboration. The Measurement of a Lepton-Lepton Electroweak Reaction (MOLLER) experiment proposes to measure the parity-violating asymmetry in electron-electron (Møller) scattering. The measurement will be carried out at Jefferson Laboratory's state-of-the-art accelerator by rapidly flipping the longitudinal polarization of electrons that have been accelerated to 11 GeV and observing the resulting fractional difference in the probability of these electrons scattering off atomic electrons in a liquid hydrogen target. This asymmetry is proportional to the weak charge of the electron, which in turn is a function of the electroweak mixing angle, a fundamental parameter of the electroweak theory. The accuracy of the proposed measurement allows for a low energy determination of the mixing angle with precision on par with the two best measurements at electron-positron colliders.

My primary responsibilities include the characterization and optimization of the Cherenkov detectors, proposed as the primary detector to measure the Møller scattered electrons. The prototype detectors have been tested at the MAMI accelerator facility in MAINZ, Germany with electron beam of ~855 MeV. I was one of the responsible personnel for commissioning the detector segment (containing eight detector prototypes) in the beamline, take data taking shifts and analysing the data. The results from the beam test has been presented by me at the PISA 2024 conference and also published as an article reporting that the prototype detectors performance satisfies the experimental requirements. The same detector prototypes are also being tested with cosmic muons at UMass to benchmark the performance with the electron beam. I am involved in the group of designing and setting up the cosmic test stand at UMass involving the GEM chambers to facilitate tracking of the muons. Currently I am working on the GEM (APV25 card based) readout using the MPD and VME to setup a tracking setup using three triple GEM prototypes. I am responsible for setting up a cosmic test stand to test the production modules before installing them in the experimental hall. Currently I am working on benchmarking the performance of the production modules using cosmic muon similar to that proposed for the final experiment. Moreover, I contribute to the development of analysis software for both event mode and integrating mode data and assist in developing and testing the DAQ for the MOLLER experiment. In addition to these primary responsibilities, I regularly take data-taking shifts at Jefferson Lab in Virginia, USA, and provide support in the experimental halls during the commissioning of other experiments.

PUBLICATIONS

Journals

- **Performance study of a bakelite RPC prototype built by new technique of linseed oil coating**
  A. Sen, S. Mandal, **S. Chatterjee**, S. Gope, S. Das, S. Biswas
  JINST 20 T03007 [arXiv.2408.07336]

- **Charging-up effect and uniformity study of a single mask triple GEM detector**
  S. Chatterjee, A. Sen, S. Das, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1049 (2023) 168110

- **Spark probability measurement of a single mask triple GEM detector**
  S. Chatterjee, U. Frankenfeld, C. Garabatos, J. Hehner, T. Morhardt, C. J. Schmidt, H. R. Schmidt, A. Lymanets, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 977 (2020) 164334 [arXiv:2107.00883]

- **J/ψ yields in low energy nuclear collisions at SPS and FAIR: A baseline estimation**
  S. Chatterjee, P.P.Bhaduri, S.Chattopadhyay
  Nuclear Physics A 1029 (2023) 122554

- **A new technique of linseed oil coating in bakelite RPC and the first test results**
  A. Sen, **S. Chatterjee**, S. Das, S. K. Ghosh, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1024 (2022) 166095 [arXiv:2108.06923]

- **Commissioning and testing of pre-series triple GEM prototypes for CBM MuCh in the mCBM experiment at the SIS18 facility of GSI**
  A. Kumar, A. Agarwal, **S. Chatterjee**, S. Chattopadhyay, A.K. Dubey, C. Ghosh, E. Nandy, V. Negi, S. K. Prasad, J. Saini, V. Singhal, O. Singh, G. Sikder, J. de Cuveland, I. Deppner, D. Emschermann, V. Friese, J. Frühauf, M. Gumiński, N. Herrmann, D. Hutter, M. Kis, J. Lehnert, P. A. Loizeau, C.J. Schmidt, C. Sturm, F. Uhlig and W. Zabołotny
  Journal of Instrumentation 16 (2021) P09002

- **Development and implementation of a time-based signal generation scheme for the Muon Chamber simulation of the CBM experiment at FAIR**
  V. Singhal, **S. Chatterjee**, V. Frise, S. Chattopadhyay
  Journal of Instrumentation 16 (2021) P08043 [arXiv:2104.12012]

- **Cosmic ray flux and lockdown due to COVID19 in Kolkata – any correlation?**
  A. Sen, **S. Chatterjee**, S. Roy, R. Biswas, S. Das, S. K. Ghosh, S. Biswas
  Pramana – J. Phys. 95, (2021) 64 [arXiv:2010.06648]

- **Stability study and time resolution measurement of Straw Tube detectors**
  S. Roy, S. Jaiswal, **S. Chatterjee**, A. Sen, S. Das, S. K. Ghosh, S. Raha, V. M. Lysan, G. D. Kekelidze, V. V. Myalkovsky, S. Biswas
  Pramana – J. Phys. 95, (2021) 50 [arXiv:2007.12547]

- **Study of charging-up effect for a single mask triple GEM detector**
  S. Chatterjee, A. Sen, S. Das, S. K. Ghosh, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1014 (2021) 165749 [arXiv:2107.00890]

- **Study of charging up effect in a triple GEM detector**
  S. Chatterjee, A. Sen, S. Roy, K. Nivedita G, A. Paul, S. Das, S. Biswas
  Journal of Instrumentation 15 (2020) T09011 [arXiv:2007.11444]

Proceedings in Journals

- **Characterization of Cherenkov detectors for the MOLLER experiment**
  S. Chatterjee (on behalf of the MOLLER collaboration)
  Nuclear Inst. and Methods in Physics Research, A 1070 (2025) 170057

- **Investigation of the stability in the performance of triple GEM detectors for High Energy Physics experiments GEM chamber**
  S. Mandal, **S. Chatterjee**, A. Sen, S. Gope, S. Dhani, A. C. Hegde, M. Chatterjee, S. Das, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1064 (2024) 169389

- **Observation on the bias current variation of a single mask triple GEM chamber**
  S. Chatterjee, A. Sen, R. Paul, S. Sahai, S. Das, S. Biswas
  JINST 18 C05002

- **Effect of relative humidity on the long-term operation of a single mask triple GEM chamber**
  S. Chatterjee, A. Sen, S. Das, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1046 (2023) 167747[arXiv:2206.04259]

- **Visual investigation of possible degradation in GEM foil under test**
  S. Chatterjee, A. Sen, S. Das, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1045 (2023) 167573 [arXiv:2206.04052]

- **Characterisation of a new RPC prototype using conventional gas mixture**
  A. Sen, **S. Chatterjee**, S. Das, S. Biswas
  Nuclear Inst. and Methods in Physics Research, A 1045 (2023) 167572 [arXiv:2206.04259]

- **Interplay between eclipses and soft cosmic rays**
  S. Roy, **S. Chatterjee**, S. Chakraborty, S. Biswas, S. Das, S. K. Ghosh, S. K. Gupta, A. Jain, I. Mazumdar, P. K. Nayak and S. Raha
  Proceedings of Science (ICRC2021) 131

- **Testing of triple GEM prototypes for the CBM Muon Chamber system in the mCBM experiment at the SIS18 facility of GSI**
  A. Kumar, C. Ghosh, **S. Chatterjee**, G. Sikder, A. K. Dubey, J. Saini, E. Nandy, V. Singhal, V. S. Negi, S. Chattopadhyay, and S.K. Prasad
  Journal of Instrumentation 15 (2020) C10020

- **Characterisation of an RPC prototype with moderate resistivity plates using tetrafluoroethane ($C_2H_2F_4$)**
  A. Sen, **S. Chatterjee**, S. Roy, S. Biswas, S. Das
  Journal of Instrumentation 15 (2020) C06055 [arXiv:2004.05469]

- **Long term stability study of triple GEM detector using different Argon based gas mixtures: an update**
  S. Chatterjee, S. Roy, A. Sen, S. Chakraborty, S. Biswas, S. Das, S. K. Ghosh, S. K. Prasad and S. Raha
  Journal of Physics: Conference Series, 1498 (2020) 012037 [arXiv:2107.00858]

- **Plastic scintillator detector array for detection of cosmic ray air shower.**
  S. Roy, S. Chakraborty **S. Chatterjee**, S. Biswas, S. Das, S. K. Ghosh, A. Maulik, S. Raha
  Nuclear Inst. and Methods in Physics Research, A 936 (2019) 249 [arXiv:1807.04951]

- **Stability study of gain and energy resolution for GEM detector.**
  S. Roy, S. Rudra, S. Shaw, **S. Chatterjee**, S. Chakraborty, R. P. Adak, S. Biswas, S. Das, S. K. Ghosh, S. K. Prasad, S. Raha
  Nuclear Inst. and Methods in Physics Research, A 936 (2019) 485

- **A new type of RPC with very low resistive material**
  S. Chakraborty, **S. Chatterjee**, S. Roy, A. Roy, S. Biswas, S. Das, S. K. Ghosh, S. K. Prasad, S. Raha
  Nuclear Inst. and Methods in Physics Research, A 936 (2019) 424 [arXiv:1807.04961]

- **Study of uniformity of characteristics over the surface for triple GEM detector**
  S. Chatterjee, S. Chakraborty, S. Roy, S. Biswas, S. Das, S.K. Ghosh, S.K. Prasad, S. Raha
  Nuclear Inst. and Methods in Physics Research, A 936 (2019) 491 [arXiv:1807.04961]

National conference proceedings

- **Performance Studies of Single Mask Triple GEM Detector for Future Heavy-Ion Experiments**
  S. Chatterjee, A. Sen, S. Das, S. Biswas
  Proceedings of the XXV DAE-BRNS High Energy Physics Symposium 2022, 563; DOI: https://doi.org/10.1007/978-981-97-0289-3

- **Effect of bias current variation on the gain of a single mask triple GEM chamber**
  S. Sahai, R. Paul, P. Barik, A. Dandapat, **S. Chatterjee**, A. Sen, S. Das, S. Biswas
  Proceedings of the DAE-BRNS Symposium on Nucl. Phys. Vol. 66, (2022) 1186

- **J/ψ production cross-sections in low energy collisions at SPS and FAIR: a data driven approach**
  S. Chatterjee, P. P. Bhaduri, S. Chattopadhyay
  Proceedings of the DAE-BRNS Symposium on Nucl. Phys. Vol. 66, (2022) 1030

- **Feasibility Studies of J/ψ Measurement with CBM Detector Setup at FAIR SIS100 Energies**
  S. Chatterjee for the CBM Collaboration
  Proceedings of the XXIV DAE-BRNS High Energy Physics, Vol. 277, (2022) 483

- **Study of charging up effect in GEM detector**
  K. Nivedita G, A. Paul, **S. Chatterjee**, S. Roy, A. Sen, S. Biswas, S. Das
  Proceedings of the DAE-BRNS Symposium on Nucl. Phys. Vol. 64, (2019) 976

- **R&D of Straw Tube detector for High Energy Physics experiments**
  S. Roy, S. Jaiswal, **S. Chatterjee**, A. Sen, S. Biswas, S. Das, S. K. Ghosh, S. Raha, V. M. Lysan, G. D. Kekelidze, V. V. Myalkovsky
  Proceedings of the DAE-BRNS Symposium on Nucl. Phys. Vol. 64, (2019) 990

- **Development and issues of bakelite RPC**
  A. Sen, R. Banerjee, A. Roy, **S. Chatterjee**, S. Roy, S. Biswas, S. Das
  Proceedings of the DAE-BRNS Symposium on Nucl. Phys. Vol. 64, (2019) 996

- **Study of cosmic ray with plastic scintillator detector.**
  S. Shaw, N. Nandi, **S. Chatterjee**, P. Chawla, S. Roy, R. P. Adak, S. Biswas, S. Das, S. K. Ghosh, S. Raha
  Proceeding of the DAE symposium on Nuclear Physics. Volume 62, (2017), 1030.

Collaboration/Internal notes

- **Estimation of J/ψ production cross-section in p+A and A+A collisions at FAIR and SPS energies**
  S. Chatterjee, P. P. Bhaduri, S. Chattopadhyay
  CBM Progress Report 2022, 166; DOI:10.15120/GSI-2023-00384

- **Study of uniformity for a single mask triple GEM detector**
  S. Chatterjee, A. Sen, S. Das, and S. Biswas
  CBM Progress Report 2022, 100; DOI:10.15120/GSI-2023-00384

- **Charging-up effect in single mask triple GEM chamber**
  S. Chatterjee, A. Sen, S. Das, S. K. Ghosh, and S. Biswas
  CBM Progress Report 2021, 99; DOI:10.15120/GSI-2022-00599

- **Reconstruction of low mass dimuon cocktail at top SIS100 energy**
  S. Chatterjee, P. P. Bhaduri, S. Chattopadhyay
  CBM Progress Report 2021, 215; DOI:10.15120/GSI-2022-00599

- **Di-muon combinatorial background estimation using different techniques**
  S. Chatterjee, P. P. Bhaduri, S. Chattopadhyay
  CBM Progress Report 2021, 218; DOI:10.15120/GSI-2022-00599

- **Estimation of J/ψ production cross-section in 30 GeV p+A collision at CBM SIS100**
  S. Chatterjee, P. P. Bhaduri, S. Chattopadhyay **CBM Progress Report 2020, 190; DOI:10.15120/GSI-2021-00421**

- **Generation of input signals for dilepton simulation using Pluto event generator**
  S. Chatterjee, T. Galatyuk, P. P. Bhaduri
  CBM Progress Report 2020, 193; DOI:10.15120/GSI-2021-00421

- **Reconstruction of di-muon cocktail using GEANT3 & GEANT4**
  S. Chatterjee, A. Senger, P. P. Bhaduri, S. Chattopadhyay
  CBM Progress Report 2020, 197; DOI:10.15120/GSI-2021-00421

- **Non-monolithic design of the 5th MuCh absorber parameters and tolerances**
  S. Chatterjee, P. P. Bhaduri, O. Singh, V. Nikulin, S. Chattopadhyay **CBM-TN-20006**

- **Effect of absorbers surface tolerance on the Muon Chamber (MuCh) performance for the CBM experiment at FAIR**
  S. Chatterjee, O. Singh, P. P. Bhaduri, S. Chattopadhyay, A. Senger and V. Nikulin
  CBM Progress Report 2019, 97; DOI: 10.15120/GSI-2020-00904

- **Effect of gaps on the fifth absorber of Muon Chamber (MuCh) for the CBM experiment at FAIR**
  S. Chatterjee, O. Singh, P. P. Bhaduri, S. Chattopadhyay, and V. Nikulin
  CBM Progress Report 2019, 99; DOI: 10.15120/GSI-2020-00904

- **Reconstruction of J/ψ mesons at SIS100 energies with realistic MuCh set up**
  S. Chatterjee, O. Singh, A. Senger, P. P. Bhaduri, and S. Chattopadhyay
  CBM Progress Report 2019, 181; DOI: 10.15120/GSI-2020-00904

- **Stability and uniformity study of triple GEM detector**
  S. Chatterjee, S. Roy, S. Chakraborty, S. Rudra, S. Shaw, R.P. Adak, S. Biswas, S. Das, S.K. Ghosh, S.K. Prasad & S. Raha
  CBM Progress Report 2018, 65; DOI:10.15120/GSI-2019-01018

Teaching experience

- **First year seminar course at the University of Massachusetts, Amherst**
  Title of the course: Exploring the Invisible: The Wonders of Subatomic Particles
  Course credit: 1
  Course duration: September - December 2024
  Course level: First year undergraduate students

Professional service

- **Editorial board member:** Radiation Science and Technology, Science Publishing Group, DOI: 10.11648/j.rst
- **Reviewer:** Measurement, Nuclear Instrumentation and Methods in Physics Research, A, Radiation Physics and Chemistry

Other activities

- Visited Jefferson Lab, USA to take data taking shift as the Target Operator and Shift Leader for the SBS GEp-V experiment at Hall A, May 2025

- Organising committee member of the UMass International Faculty & Staff Network group, 2024-2025

- Visited Jefferson Lab, USA to take data taking shift as the Target Operator for the SBS GEn-II experiment at Hall A, 25.02.2022 - 03.03.2023

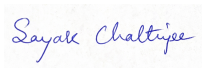
- Judge at the Senior School Science Exhibition at La Martiniere for Boys, Kolkata, India, 2022

- Quiz master at the FAIR week of Vigyan Samagam in Mumbai, Bengaluru and Kolkata, India, September 2019

- FAIR project volunteer at Vigyan Samagam in Mumbai, Bengaluru and Kolkata, India, September 2019

- Member of the organizing committee for 34th CBM Collaboration meeting, at Bose Institute, Kolkata, India, September 2019

- Member of the organizing committee for the workshop on Forward and Jet physics at LHC, Bose Institute, Kolkata, India, February 2018



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